

sg13g2_io_fast_1p65V_3p6V_m40C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx



*sg13g2_io_fast_1p65V_3p6V_m40C Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C, Voltage 1.65, Temp
-40.00*

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.03154	0.02880	0.29002	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.03154	0.02880	0.37334	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.03154	0.02880	0.21785	0.24000	12.00000
sg13g2_IOPadOut16mA	0.04388	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.04388	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.04387	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	110.67000	9432.99217	22152.20000
sg13g2_IOPadInOut30mA	216.84700	9727.89511	22153.10000
sg13g2_IOPadInOut4mA	138.69400	9243.27483	22152.40000
sg13g2_IOPadOut16mA	360.16300	2443.32583	6375.78000
sg13g2_IOPadOut30mA	554.84300	2729.70883	6374.21000
sg13g2_IOPadOut4mA	256.88200	2227.79717	6375.53000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (RR)	0.12000	0.02400	0.16420	0.80000	0.14400	0.28583	5.00000	0.24000	0.48829
sg13g2_IOPadInOut30mA	pad->p2c (RR)	0.12000	0.02400	0.16408	0.80000	0.14400	0.28596	5.00000	0.24000	0.48837
sg13g2_IOPadInOut4mA	pad->p2c (RR)	0.12000	0.02400	0.16420	0.80000	0.14400	0.28569	5.00000	0.24000	0.48800

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (FF)	0.12000	0.02400	0.16646	0.80000	0.14400	0.30428	5.00000	0.24000	0.56576
sg13g2_IOPadInOut30mA	pad->p2c (FF)	0.12000	0.02400	0.16651	0.80000	0.14400	0.30445	5.00000	0.24000	0.56699
sg13g2_IOPadInOut4mA	pad->p2c (FF)	0.12000	0.02400	0.16653	0.80000	0.14400	0.30407	5.00000	0.24000	0.56470

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (RR)	0.02000	1.00000	1.04218	0.33000	9.00000	1.75530	2.50000	20.00000	2.59531
	c2p_en->pad (FR)	0.02000	1.00000	0.75457	0.33000	9.00000	0.79687	2.50000	20.00000	0.93773
	c2p_en->pad (RR)	0.02000	1.00000	1.03594	0.33000	9.00000	1.79290	2.50000	20.00000	2.69472
sg13g2_IOPadInOut30mA	c2p->pad (RR)	0.02000	2.00000	1.31277	0.33000	18.00000	2.19615	2.50000	40.00000	3.10997
	c2p_en->pad (FR)	0.02000	2.00000	0.91356	0.33000	18.00000	0.95542	2.50000	40.00000	1.09595
	c2p_en->pad (RR)	0.02000	2.00000	1.29765	0.33000	18.00000	2.23661	2.50000	40.00000	3.21449
sg13g2_IOPadInOut4mA	c2p->pad (RR)	0.02000	1.00000	1.03321	0.33000	5.00000	2.26070	2.50000	12.00000	4.37521
	c2p_en->pad (FR)	0.02000	1.00000	0.59975	0.33000	5.00000	0.63965	2.50000	12.00000	0.78069
	c2p_en->pad (RR)	0.02000	1.00000	1.03637	0.33000	5.00000	2.31707	2.50000	12.00000	4.52944
sg13g2_IOPadOut16mA	c2p->pad (RR)	0.02000	1.00000	1.00786	0.33000	9.00000	1.78360	2.50000	20.00000	3.00117
sg13g2_IOPadOut30mA	c2p->pad (RR)	0.02000	2.00000	1.28045	0.33000	18.00000	2.22319	2.50000	40.00000	3.51851
sg13g2_IOPadOut4mA	c2p->pad (RR)	0.02000	1.00000	0.99164	0.33000	5.00000	2.28154	2.50000	12.00000	4.77129

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	0.71186	0.33000	9.00000	1.37369	2.50000	20.00000	2.43268
	c2p_en->pad (FF)	0.02000	1.00000	0.71753	0.33000	9.00000	0.76245	2.50000	20.00000	0.95301
	c2p_en->pad (RF)	0.02000	1.00000	0.57676	0.33000	9.00000	1.30879	2.50000	20.00000	2.22893
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	0.97853	0.33000	18.00000	1.72575	2.50000	40.00000	2.80943
	c2p_en->pad (FF)	0.02000	2.00000	1.15157	0.33000	18.00000	1.19282	2.50000	40.00000	1.38122
	c2p_en->pad (RF)	0.02000	2.00000	0.69813	0.33000	18.00000	1.53689	2.50000	40.00000	2.52671
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	0.74968	0.33000	5.00000	2.03018	2.50000	12.00000	4.42297
	c2p_en->pad (FF)	0.02000	1.00000	0.35078	0.33000	5.00000	0.40480	2.50000	12.00000	0.58667
	c2p_en->pad (RF)	0.02000	1.00000	0.76263	0.33000	5.00000	2.12536	2.50000	12.00000	4.47172
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	0.67202	0.33000	9.00000	1.30560	2.50000	20.00000	2.22235
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	0.94034	0.33000	18.00000	1.66247	2.50000	40.00000	2.60365
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	0.69542	0.33000	5.00000	1.95040	2.50000	12.00000	4.19531

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	-1.41270	0.80000	0.14400	-1.17090	5.00000	0.24000	0.52363
	pad	0.12000	0.02400	0.05156	0.80000	0.14400	0.05252	5.00000	0.24000	0.06594
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	-2.21757	0.80000	0.14400	-1.98208	5.00000	0.24000	-0.28346
	pad	0.12000	0.02400	0.05169	0.80000	0.14400	0.05286	5.00000	0.24000	0.06631
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	-0.75982	0.80000	0.14400	-0.51178	5.00000	0.24000	1.18159
	pad	0.12000	0.02400	0.05124	0.80000	0.14400	0.05231	5.00000	0.24000	0.06565

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	2.02351	0.80000	0.14400	2.28100	5.00000	0.24000	3.98231
	pad	0.12000	0.02400	0.13216	0.80000	0.14400	0.15374	5.00000	0.24000	0.23984
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	2.83554	0.80000	0.14400	3.08854	5.00000	0.24000	4.80379
	pad	0.12000	0.02400	0.13185	0.80000	0.14400	0.15332	5.00000	0.24000	0.23865
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	1.36374	0.80000	0.14400	1.62192	5.00000	0.24000	3.32831
	pad	0.12000	0.02400	0.13234	0.80000	0.14400	0.15393	5.00000	0.24000	0.24121

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	8.21427	0.33000	9.00000	8.05057	2.50000	20.00000	7.96576
	c2p	0.02000	1.00000	2.83295	0.33000	9.00000	24.33120	2.50000	20.00000	47.32960
	c2p_en	0.02000	1.00000	7.59670	0.33000	9.00000	10.33150	2.50000	20.00000	13.63310
	c2p_en	0.02000	1.00000	2.77463	0.33000	9.00000	24.29420	2.50000	20.00000	47.93130
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	13.07250	0.33000	18.00000	11.95980	2.50000	40.00000	10.80840
	c2p	0.02000	2.00000	5.51366	0.33000	18.00000	47.96840	2.50000	40.00000	32.86640
	c2p_en	0.02000	2.00000	11.67100	0.33000	18.00000	16.81560	2.50000	40.00000	22.75310
	c2p_en	0.02000	2.00000	5.45806	0.33000	18.00000	48.00370	2.50000	40.00000	32.73480
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	4.65603	0.33000	5.00000	5.21293	2.50000	12.00000	6.21750
	c2p	0.02000	1.00000	2.83798	0.33000	5.00000	13.66860	2.50000	12.00000	33.28940
	c2p_en	0.02000	1.00000	4.34056	0.33000	5.00000	5.90096	2.50000	12.00000	9.12432
	c2p_en	0.02000	1.00000	2.77980	0.33000	5.00000	13.62590	2.50000	12.00000	33.39960
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	7.69341	0.33000	9.00000	7.36265	2.50000	20.00000	7.16368
	c2p	0.02000	1.00000	-0.03646	0.33000	9.00000	0.05537	2.50000	20.00000	0.81014
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	12.73040	0.33000	18.00000	11.56490	2.50000	40.00000	11.19020
	c2p	0.02000	2.00000	-0.03646	0.33000	18.00000	0.05538	2.50000	40.00000	0.81407
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	3.89220	0.33000	5.00000	3.78230	2.50000	12.00000	3.82712
	c2p	0.02000	1.00000	-0.03646	0.33000	5.00000	0.05538	2.50000	12.00000	0.81457

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	19.70850	0.33000	9.00000	12.80900	2.50000	20.00000	10.54620
	c2p	0.02000	1.00000	0.38047	0.33000	9.00000	0.48744	2.50000	20.00000	1.29121
	c2p_en	0.02000	1.00000	4.28730	0.33000	9.00000	4.68143	2.50000	20.00000	5.27787
	c2p_en	0.02000	1.00000	0.20575	0.33000	9.00000	0.28295	2.50000	20.00000	0.74400
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	70.52100	0.33000	18.00000	43.08410	2.50000	40.00000	31.20430
	c2p	0.02000	2.00000	0.38587	0.33000	18.00000	0.48874	2.50000	40.00000	1.27905
	c2p_en	0.02000	2.00000	6.51917	0.33000	18.00000	6.93699	2.50000	40.00000	7.56408
	c2p_en	0.02000	2.00000	0.20711	0.33000	18.00000	0.28456	2.50000	40.00000	0.73210
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	3.38178	0.33000	5.00000	4.21326	2.50000	12.00000	5.74316
	c2p	0.02000	1.00000	0.39312	0.33000	5.00000	0.52100	2.50000	12.00000	1.35679
	c2p_en	0.02000	1.00000	2.59839	0.33000	5.00000	3.46480	2.50000	12.00000	4.97523
	c2p_en	0.02000	1.00000	0.22017	0.33000	5.00000	0.31750	2.50000	12.00000	0.81248
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	20.38220	0.33000	9.00000	12.27620	2.50000	20.00000	9.76579
	c2p	0.02000	1.00000	0.12183	0.33000	9.00000	0.22179	2.50000	20.00000	0.97871
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	72.34420	0.33000	18.00000	43.10480	2.50000	40.00000	31.59990
	c2p	0.02000	2.00000	0.12179	0.33000	18.00000	0.22183	2.50000	40.00000	0.97867
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.81441	0.33000	5.00000	2.69314	2.50000	12.00000	2.72816
	c2p	0.02000	1.00000	0.12180	0.33000	5.00000	0.22174	2.50000	12.00000	0.99882

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	0.02000	-0.03802	0.33000	-0.03990	2.50000	-0.04019
sg13g2_IOPadInOut30mA	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	0.02000	-0.03802	0.33000	-0.03990	2.50000	-0.04019
sg13g2_IOPadInOut4mA	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	0.02000	-0.03803	0.33000	-0.03990	2.50000	-0.04019

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	0.02000	0.04816	0.33000	0.04753	2.50000	0.04736
sg13g2_IOPadInOut30mA	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	0.02000	0.04818	0.33000	0.04771	2.50000	0.04736
sg13g2_IOPadInOut4mA	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	0.02000	0.04820	0.33000	0.04783	2.50000	0.04736

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * pad * p2c)	0.02000	-0.03802	0.33000	-0.03990	2.50000	-0.04019
	(!c2p_en * !pad * !p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * !pad * !p2c)	0.02000	-0.03802	0.33000	-0.03990	2.50000	-0.04019
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * pad * p2c)	0.02000	-0.03803	0.33000	-0.03990	2.50000	-0.04019
	(!c2p_en * !pad * !p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * !pad * !p2c)	0.02000	-0.03802	0.33000	-0.03990	2.50000	-0.04019
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * pad * p2c)	0.02000	-0.03803	0.33000	-0.03990	2.50000	-0.04019
	(!c2p_en * !pad * !p2c)	0.02000	-0.00012	0.33000	-0.00010	2.50000	-0.00010
	(!c2p_en * !pad * !p2c)	0.02000	-0.03803	0.33000	-0.03990	2.50000	-0.04019

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * pad * p2c)	0.02000	0.04816	0.33000	0.04755	2.50000	0.04736
	(!c2p_en * !pad * !p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * !pad * !p2c)	0.02000	0.04816	0.33000	0.04753	2.50000	0.04736
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * pad * p2c)	0.02000	0.04818	0.33000	0.04774	2.50000	0.04736
	(!c2p_en * !pad * !p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * !pad * !p2c)	0.02000	0.04818	0.33000	0.04771	2.50000	0.04736
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * pad * p2c)	0.02000	0.04820	0.33000	0.04783	2.50000	0.04736
	(!c2p_en * !pad * !p2c)	0.02000	0.00012	0.33000	0.00010	2.50000	0.00010
	(!c2p_en * !pad * !p2c)	0.02000	0.04820	0.33000	0.04783	2.50000	0.04736

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	1.95190	0.33000	1.95366	2.50000	1.95557
	0.02000	0.06236	0.33000	0.10964	2.50000	0.55452
sg13g2_IOPadInOut30mA	0.02000	3.15931	0.33000	3.16012	2.50000	3.16710
	0.02000	0.06233	0.33000	0.10960	2.50000	0.55461
sg13g2_IOPadInOut4mA	0.02000	0.91810	0.33000	0.91883	2.50000	0.92087
	0.02000	0.06235	0.33000	0.10960	2.50000	0.55406

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.43602	0.33000	0.43188	2.50000	0.42856
	0.02000	0.16713	0.33000	0.21952	2.50000	0.66034
sg13g2_IOPadInOut30mA	0.02000	0.42199	0.33000	0.41846	2.50000	0.41478
	0.02000	0.16716	0.33000	0.21947	2.50000	0.66020
sg13g2_IOPadInOut4mA	0.02000	0.47029	0.33000	0.46595	2.50000	0.46313
	0.02000	0.16712	0.33000	0.21947	2.50000	0.66054

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	8.21427	0.33000	8.05057	2.50000	7.96576
	0.02000	2.83295	0.33000	24.33120	2.50000	47.32960
	0.02000	7.59670	0.33000	10.33150	2.50000	13.63310
	0.02000	2.77463	0.33000	24.29420	2.50000	47.93130
sg13g2_IOPadInOut30mA	0.02000	13.07250	0.33000	11.95980	2.50000	10.80840
	0.02000	5.51366	0.33000	47.96840	2.50000	32.86640
	0.02000	11.67100	0.33000	16.81560	2.50000	22.75310
	0.02000	5.45806	0.33000	48.00370	2.50000	32.73480
sg13g2_IOPadInOut4mA	0.02000	4.65603	0.33000	5.21293	2.50000	6.21750
	0.02000	2.83798	0.33000	13.66860	2.50000	33.28940
	0.02000	4.34056	0.33000	5.90096	2.50000	9.12432
	0.02000	2.77980	0.33000	13.62590	2.50000	33.39960
sg13g2_IOPadOut16mA	0.02000	7.69341	0.33000	7.36265	2.50000	7.16368
	0.02000	-0.03646	0.33000	0.05537	2.50000	0.81014
sg13g2_IOPadOut30mA	0.02000	12.73040	0.33000	11.56490	2.50000	11.19020
	0.02000	-0.03646	0.33000	0.05538	2.50000	0.81407
sg13g2_IOPadOut4mA	0.02000	3.89220	0.33000	3.78230	2.50000	3.82712
	0.02000	-0.03646	0.33000	0.05538	2.50000	0.81457

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	19.70850	0.33000	12.80900	2.50000	10.54620
	0.02000	0.38047	0.33000	0.48744	2.50000	1.29121
	0.02000	4.28730	0.33000	4.68143	2.50000	5.27787
	0.02000	0.20575	0.33000	0.28295	2.50000	0.74400
sg13g2_IOPadInOut30mA	0.02000	70.52100	0.33000	43.08410	2.50000	31.20430
	0.02000	0.38587	0.33000	0.48874	2.50000	1.27905
	0.02000	6.51917	0.33000	6.93699	2.50000	7.56408
	0.02000	0.20711	0.33000	0.28456	2.50000	0.73210
sg13g2_IOPadInOut4mA	0.02000	3.38178	0.33000	4.21326	2.50000	5.74316
	0.02000	0.39312	0.33000	0.52100	2.50000	1.35679
	0.02000	2.59839	0.33000	3.46480	2.50000	4.97523
	0.02000	0.22017	0.33000	0.31750	2.50000	0.81248
sg13g2_IOPadOut16mA	0.02000	20.38220	0.33000	12.27620	2.50000	9.76579
	0.02000	0.12183	0.33000	0.22179	2.50000	0.97871
sg13g2_IOPadOut30mA	0.02000	72.34420	0.33000	43.10480	2.50000	31.59990
	0.02000	0.12179	0.33000	0.22183	2.50000	0.97867
sg13g2_IOPadOut4mA	0.02000	2.81441	0.33000	2.69314	2.50000	2.72816
	0.02000	0.12180	0.33000	0.22174	2.50000	0.99882

INPUT



*sg13g2_io_fast_1p65V_3p6V_m40C Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C, Voltage 1.65, Temp -40.00*

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23272	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	153.25400	1105.40283	3196.40000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (RR)	0.12000	0.02400	0.16400	0.80000	0.14400	0.28580	5.00000	0.24000	0.48771

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (FF)	0.12000	0.02400	0.16621	0.80000	0.14400	0.30378	5.00000	0.24000	0.56346

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	-0.67175	0.80000	0.14400	-0.41958	5.00000	0.24000	1.26874
	pad	0.12000	0.02400	0.05185	0.80000	0.14400	0.05351	5.00000	0.24000	0.06652

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	1.27244	0.80000	0.14400	1.53353	5.00000	0.24000	3.24359
	pad	0.12000	0.02400	0.13102	0.80000	0.14400	0.15280	5.00000	0.24000	0.24044

SG13G2_CORNER



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG



sg13g2_io_fast_1p65V_3p6V_m40C

Cell Library: Process

sg13g2_io_fast_1p65V_3p6V_m40C,

Voltage 1.65, Temp -40.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.74927	0.61022	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	1359.77000	2719.54000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (R)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (F)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

SG13G2_IOPADIOVDD



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS



sg13g2_io_fast_1p65V_3p6V_m40C
Cell Library: Process
sg13g2_io_fast_1p65V_3p6V_m40C,
Voltage 1.65, Temp -40.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx



*sg13g2_io_fast_1p65V_3p6V_m40C Cell Library:
Process sg13g2_io_fast_1p65V_3p6V_m40C, Voltage
1.65, Temp -40.00*

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.03098	0.02946	20.00000
sg13g2_IOPadTriOut30mA	0.03098	0.02946	40.00000
sg13g2_IOPadTriOut4mA	0.03098	0.02945	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	185.13400	8282.39750	19894.00000
sg13g2_IOPadTriOut30mA	332.87700	8467.40200	19895.70000
sg13g2_IOPadTriOut4mA	80.12220	8132.39996	19893.60000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	1.04194	0.33000	9.00000	1.75614	2.50000	20.00000	2.59577
	c2p_en->pad (FR)	0.02000	1.00000	0.76081	0.33000	9.00000	0.80312	2.50000	20.00000	0.94695
	c2p_en->pad (RR)	0.02000	1.00000	1.03697	0.33000	9.00000	1.79487	2.50000	20.00000	2.69549
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	1.31435	0.33000	18.00000	2.19720	2.50000	40.00000	3.11280
	c2p_en->pad (FR)	0.02000	2.00000	0.92001	0.33000	18.00000	0.96105	2.50000	40.00000	1.10530
	c2p_en->pad (RR)	0.02000	2.00000	1.30051	0.33000	18.00000	2.24286	2.50000	40.00000	3.21916
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	1.02652	0.33000	5.00000	2.25365	2.50000	12.00000	4.36709
	c2p_en->pad (FR)	0.02000	1.00000	0.60720	0.33000	5.00000	0.64747	2.50000	12.00000	0.79094
	c2p_en->pad (RR)	0.02000	1.00000	1.03278	0.33000	5.00000	2.31591	2.50000	12.00000	4.52736

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	0.71499	0.33000	9.00000	1.37656	2.50000	20.00000	2.43971
	c2p_en->pad (FF)	0.02000	1.00000	0.72324	0.33000	9.00000	0.77951	2.50000	20.00000	0.96254
	c2p_en->pad (RF)	0.02000	1.00000	0.58040	0.33000	9.00000	1.31996	2.50000	20.00000	2.24898
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	0.98337	0.33000	18.00000	1.73176	2.50000	40.00000	2.81668
	c2p_en->pad (FF)	0.02000	2.00000	1.15893	0.33000	18.00000	1.19649	2.50000	40.00000	1.39170
	c2p_en->pad (RF)	0.02000	2.00000	0.70113	0.33000	18.00000	1.54251	2.50000	40.00000	2.53392
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	0.74509	0.33000	5.00000	2.02581	2.50000	12.00000	4.42046
	c2p_en->pad (FF)	0.02000	1.00000	0.35722	0.33000	5.00000	0.41140	2.50000	12.00000	0.59731
	c2p_en->pad (RF)	0.02000	1.00000	0.76026	0.33000	5.00000	2.12544	2.50000	12.00000	4.47477

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	7.65313	0.33000	9.00000	7.34821	2.50000	20.00000	6.92694
	c2p	0.02000	1.00000	0.09859	0.33000	9.00000	0.17351	2.50000	20.00000	0.96004
	c2p_en	0.02000	1.00000	6.95783	0.33000	9.00000	9.53705	2.50000	20.00000	12.81280
	c2p_en	0.02000	1.00000	0.04046	0.33000	9.00000	0.13031	2.50000	20.00000	0.92575
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	12.58440	0.33000	18.00000	11.51190	2.50000	40.00000	10.43060
	c2p	0.02000	2.00000	0.09860	0.33000	18.00000	0.17352	2.50000	40.00000	0.96027
	c2p_en	0.02000	2.00000	11.09220	0.33000	18.00000	16.92950	2.50000	40.00000	22.98310
	c2p_en	0.02000	2.00000	0.04046	0.33000	18.00000	0.13031	2.50000	40.00000	0.92718
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	3.88140	0.33000	5.00000	3.72391	2.50000	12.00000	3.49790
	c2p	0.02000	1.00000	0.09861	0.33000	5.00000	0.17350	2.50000	12.00000	0.95945
	c2p_en	0.02000	1.00000	3.56526	0.33000	5.00000	4.46360	2.50000	12.00000	6.01837
	c2p_en	0.02000	1.00000	0.04048	0.33000	5.00000	0.13029	2.50000	12.00000	0.94802

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	19.18080	0.33000	9.00000	11.91250	2.50000	20.00000	9.07795
	c2p	0.02000	1.00000	0.24100	0.33000	9.00000	0.32219	2.50000	20.00000	1.10527
	c2p_en	0.02000	1.00000	3.80292	0.33000	9.00000	3.80492	2.50000	20.00000	3.80216
	c2p_en	0.02000	1.00000	0.06889	0.33000	9.00000	0.11650	2.50000	20.00000	0.54986
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	69.75640	0.33000	18.00000	42.10220	2.50000	40.00000	29.62860
	c2p	0.02000	2.00000	0.24102	0.33000	18.00000	0.32222	2.50000	40.00000	1.10563
	c2p_en	0.02000	2.00000	6.00686	0.33000	18.00000	6.01082	2.50000	40.00000	6.01016
	c2p_en	0.02000	2.00000	0.06891	0.33000	18.00000	0.11652	2.50000	40.00000	0.55006
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.75057	0.33000	5.00000	2.71548	2.50000	12.00000	2.73191
	c2p	0.02000	1.00000	0.24122	0.33000	5.00000	0.32234	2.50000	12.00000	1.10522
	c2p_en	0.02000	1.00000	1.94381	0.33000	5.00000	1.94245	2.50000	12.00000	1.94238
	c2p_en	0.02000	1.00000	0.06905	0.33000	5.00000	0.11664	2.50000	12.00000	0.55402

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	-0.00005	0.33000	-0.00004	2.50000	-0.00004
	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029
sg13g2_IOPadTriOut30mA	0.02000	0.00004	0.33000	0.00005	2.50000	0.00005
	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029
sg13g2_IOPadTriOut4mA	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00008	0.33000	0.00007	2.50000	0.00007
	0.02000	0.04628	0.33000	0.04572	2.50000	0.04488
sg13g2_IOPadTriOut30mA	0.02000	-0.00004	0.33000	-0.00005	2.50000	-0.00005
	0.02000	0.04627	0.33000	0.04593	2.50000	0.04507
sg13g2_IOPadTriOut4mA	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	0.02000	0.04627	0.33000	0.04592	2.50000	0.04556

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	-0.00005	0.33000	-0.00004	2.50000	-0.00004
	!c2p_en	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00004	0.33000	0.00005	2.50000	0.00005
	!c2p_en	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	!c2p_en	0.02000	-0.03790	0.33000	-0.03989	2.50000	-0.04029

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00008	0.33000	0.00007	2.50000	0.00007
	!c2p_en	0.02000	0.04628	0.33000	0.04572	2.50000	0.04488
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	-0.00004	0.33000	-0.00005	2.50000	-0.00005
	!c2p_en	0.02000	0.04627	0.33000	0.04593	2.50000	0.04507
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	!c2p_en	0.02000	0.04627	0.33000	0.04592	2.50000	0.04556